

BitZ logarithmic

The goal of this note is to modify BitZ-PCS so that the verifier and proof size are logarithmic.

The current obstacle right now is that in Step 4 the prover sends $\ell_2 = O(\sqrt{\ell})$ integers $\mu_1, \dots, \mu_{\ell_2}$ (we assume we are in the basic setting of Section 2.1 (Warm-up: the prime field case) of the BitZ paper), and $\mathbf{V}$ checks that

$$\sum_{j \in [\ell_2]} \pi_q(\mu_j) \cdot \mathbf{u}_j^{(2)} = \mu \quad \text{in } \mathbb{F}_q,$$

and moreover the grand product uses the values $g^{\mu_1}, \dots, g^{\mu_{\ell_2}}$, which are computed by the verifier.

Additionally, at the end of BitZ-PCS, the verifier computes an MLE eval of

$$\left(g^{\pi_q^{-1}(\mathbf{u}_i^{(1)})} \right)_{i \in [\ell_1]}. \quad (\star)$$

Let's leave $(\star)$ for later. I want to propose a solution for the first issue.

The solution has $\mathbf{P}$ commit to

$$\text{bits}(\mu_1, \dots, \mu_{\ell_2})$$

with BitZ. Then the claim

$$\sum_j \pi_q(\mu_j) \cdot \mathbf{u}_j^{(2)} = \mu \quad \text{in } \mathbb{F}_q$$

is proved with BitZ.

The grand products to be proved in standard BitZ are

$$g^{\mu_j} = \prod_i \left[\left(g^{\pi_q^{-1}(\mathbf{u}_i^{(1)})} - 1 \right) \cdot \pi_2(\text{bits}(\mathbf{f})_{ij}) + 1 \right].$$

We replace the left-hand side by

$$g^{\mu_j} = \prod_k g^{\mu_{jk} \cdot 2^{k-1}} = \prod_k ((g-1) \cdot \mu_{jk} + 1)^{2^{k-1}},$$

where $\mu_{jk} \in \{0, 1\}$ for all j, k , and k ranges from 1 to $\log(\max_j(\mu_j))$.

We use a batched GKR protocol to prove

$$\prod_k ((g-1) \cdot \mu_{jk} + 1)^{2^{k-1}} = \prod_i \left[\left(g^{\pi_q^{-1}(\mathbf{u}_i^{(1)})} - 1 \right) \cdot \pi_2(\text{bits}(\mathbf{f})_{ij}) + 1 \right]$$

for all j .

At the end we have MLE claims for (μ_{j*}) and $(\text{bits}(\mathbf{f})_{*j})$ and $\left(g^{\pi_q^{-1}(\mathbf{u}_i^{(1)})} \right)_i$.

The prover proves the first two with Ligerito.

Note that, when proving

$$\sum_j \pi_q(\mu_j) \cdot \mathbf{u}_j^{(2)} = \mu \quad \text{in } \mathbb{F}_q$$

with the recursive call to $\mathbb{Bit}\mathbb{Z}$, at the end of GKR, $\mathbf{P}$ and $\mathbf{V}$ have a MLE claim for $\mathbf{bits}(\mu_1, \dots, \mu_{\ell_2})$.

This claim as well as those for (μ_{j*}) , $(\mathbf{bits}(\mathbf{f})_{*j})$, can all be batched together in a single run of Ligerito/WHIR.

It remains to see how to avoid $\mathbf{V}$ having to check the MLE claim on

$$\left(g^{\pi_q^{-1}(\mathbf{u}_i^{(1)})}\right)_i$$

by itself. For now, let's have it check it itself.

In the recursive call of $\mathbb{Bit}\mathbb{Z}$ for the claims

$$\sum_j \pi_q(\mu_j) \cdot \mathbf{u}_j^{(2)} = \mu \quad \text{in } \mathbb{F}_q,$$

we could use the same strategy, to further save on prover time and proof size. But for now let's have only one recursive call.